
(a) Query-level certification protocol
Declared inputs
layout sensitivity $\mathbf{S}_L$
noise covariance Σ
prior box $|\theta_i| \le b$
discrepancy bound ρ
query $\mathbf{q}$ (scalar $\mathbf{q}^\top\theta$)
publish each scalar with its certificate
Level 1 rank: is q estimable at all?
$\mathbf{q} \in \mathrm{row}(\mathbf{S}_L)$; rank floor $b\, d_1(\mathbf{q}, \mathrm{row}\,\mathbf{S}_L)$
Level 2 noise: which directions pass screening?
Fisher split at $\lambda^* = (z/b)^2$; $p_{\mathrm{null}}(\mathbf{q}) \to$ state
Level 3 certified: how precise is the interval?
$w^* = \min_{\mathbf{a}}\ z\|\mathbf{a}\|_2 + b\|\tilde{\mathbf{S}}^\top\mathbf{a} - \mathbf{q}\|_1 + \tilde{\rho}^\top|\mathbf{a}|$
budget = noise + aliasing + discrepancy
REPORT
estimate $\pm\, w^*$
state $p_{\mathrm{null}} \le \tau_1$
BOUND
estimate $\pm\, w^*$
$\tau_1 < p_{\mathrm{null}} \le \tau_2$
REFUSE
prior interval
+ limiting term
(b) Break-even criterion
half-width / prior b
noise-only half-width / b
certified w^*/b
unit eigenvalue floor ($\lambda = 1$): half-width 19.6 b
prior wins
measurement wins
$\lambda^* = (z/b)^2$
Fisher eigenvalue λ along a basis query (b = 0.1)
(c) Error budget of three field queries
noise
aliasing
discrepancy
rank floor
mode 3, exact
mode 3, monthly
mode 12, exact
mode 12, monthly
mode 9, exact
mode 9, monthly
0.17
0.53
0.15
prior (zero estimator)
certified half-width w^*/w_0 (KW51 full array, b/σ = 50)